Task 1:- Basic Conditional statements and Looping programs.

a) Give 5 members, how many are even or odd

Aim:- To design an algorithm and java program that takes 5 numbers as input and determines how many of them are even and how many are odd.

Algorithm:-

step-1:- Start

step-2:- Initialize two counters

step-3:- Read 5 number from the user

step-4:- For each number
  * If the number % 2 == 0 → increment evencount
  * else → increament odd count

step-5:- Display evencount and odd count.

step-6:- Stop.

Programs-

```
import java util.scanner;
public class Even Odd counter
Public static void Main (string [ ]args) {
  Scanner sc = new scanner (system.in);
    int even count = 0, odd count = 0;
      int [ ] numbers = new int [5];
  System out . print ln ("Enter 5 numbers:");
      for (int i=0; i<5; i++) {
        number [i] % 2 == 0) {
        } else {
```

```
}
}
System.out println ("Total Even Number:" Event count);
System .out println ("Total odd Number:" Odd count);
SC close ( ) ;
}
}
```

Input:-

Enter 5 numbers:

12, 15, 9, 4, 3

Output:-

Total even number: 2

Total odd number: 3

b) Sum of last digit of two given numbers.

Aim:- To write a Program that finds the sum of two given numbers.

Algorithm:-

1) Start

2) Input two numbers, say a and b.

3) Find the last digit of "a" using a%10

4) Find the last digit of "b" using b%10

5) Add these two last digits.

6) output the result.

7) Stop

Program:-

```
import java.util.scanner;
Public class last digit sum{
Public static ~~last digit s~~ void main (string[] args) {
  Scanner sc = new scanner (system.in)
  System.out.print ("Enter first number:");
      int a = sc.next Int();
  System.out.print ("Enter pri number:");
      int b = sc.next Int();
      int last digit A = a%10;
      int last digit B = b%10;
      int ~~last di~~ sum = last digit A + last digit B;
  System.out.print ln ("sum of last digit = " + )
  sc.close();
  }
  }
```

Result:-
The given program successfully verified and executed

Input:-

Enter first number:22

Enter ~~for~~ second number:28

Output:-

Sum of last digits =10

C) To check wheather a given number is prime.

Aim:- To design and implement a java program that determines whether a user inputed integer is a prime numbers.

Algorithm:-

step-1:- Start

step-2:- Initialize an integer variable flag initialized to 1.

Step-3:- Read an integger number from the user

step-4:- If number <=1, set flag=0

step-5:- If n>1, start a loop with a counter: begin at 2.

step6:- If n% i==0

step7:- if flag ==1 display Is prime

step 8:- stop

Program:-

```
import, java.until.scanner;
 class Is prime {
 public static void main (string[] args) {
 Scanner SC = newScanner (System.in);
 System.out.print ("Enter a number");
 int num =sc.next Int();
  int flag = 1;
  if (num <=1) {
    flag =0;
  } else {
   for (int i =2; i<= num /2; i++) {
        if (num % i= =0) {
             flag =0;
             break;
            }
            }
```

```
}
if(flag==1) {
System.out.println("Prime number");
    } else {
    System out.println ("Prime number");
    }
    }
    }
```

Input :

enter a number : 66

Output :-

Not a prime number

D. Factorial of a number

Aim:- To calculate the factorial of a given number which positive integer using an iterative approach.

Algorithms:-

Step-1:- Start

Step-2:- Read an integer num from the user

Step-3:- set fact =1.

step-4:- initialize i=1 to i<=num

->multiply fact by i and get the fact values with new value

Step-5:- After the loops ends, that contains the factorial of number print the result.

Step-6:- Stop

Program:-

```
import java.util.scanner;
class-Factorial {
public static void main (string [] args) {
Scanner sc = new Scanner (system.in);
System..Out print ("enter a number:" );
    int num=sc.nxtInt();
     int fact =1;
  for(int i =1; i<=num, i++) {
   fact =fact =fact*i;
   }
   System out println (" Factorial of "+num+" is:"+fact)
   }
}
```

Input:-

enter a number:42

Output:-

factorial of 42 is :0

e) N^{th} fibonacci

Aim:- To develop a java program that reads an integer is from the users and prints the Nth fibonacci number using an iterative approach.

Algorithm :-

Step-1:- Start

Step-2:- The user to enter the value of n

Step-3:- Initialize variables a=0, b=1, c.

step-4:- If n=1 the output a
If n=2 the output b.

step-5:- Repeat from i=ton, c=a+b, a=b, b=c

Step-6:- STOP

Program:-

```
import java.util.scanner;
Class Nth fibonacci {
public static void main(string [] args){
Scanner sc = new scanner (system in);
System.out.print ("Enter a value of N:");
  int n =sc.next Int();
  int a=0, b=1, c;
    if (n==1){
System.out println ("Nth fibonaccin number is"+a);
 } else {
  for (int i=3, i<=n; i++){
     c=a+b;
      a=b;
      b=c
      }
```

```
system.out.println ("Nth fibonacci number is; "+ b);
    }
    }
```

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Result:- Hence the java program was successfully executed.

Input:-

enter a number n=56

Output:-

nth fibonacci. number is : 2144908973